

Thyroid Disease Classification Using Machine Learning

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CP322 – Machine Learning

April 2026

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Abstract

This project investigates the application of machine learning techniques for the classification of thyroid diseases using the publicly available UCI Thyroid Disease dataset. The dataset contains over 9,000 records with multiple clinical and demographic features, posing a multi-class classification problem with significant class imbalance. Three models, Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest were implemented and evaluated using accuracy, macro F1 score, and confusion matrices. While Random Forest achieved the highest overall accuracy, further analysis revealed a strong bias toward the majority class, leading to poor performance on minority classes. This highlights the limitations of relying solely on accuracy in imbalanced medical datasets. The results emphasize the importance of appropriate evaluation metrics and the need for techniques such as class balancing or re-sampling to improve model fairness and reliability. Overall, this study demonstrates both the potential and challenges of applying machine learning in healthcare classification tasks.

1. Introduction

1.1 Overview

Thyroid disorders affect essential body functions such as metabolism, heart rate, and energy regulation. Because of this, accurate and timely diagnosis is important for effective treatment and long term patient health. However, thyroid diagnosis is not always straightforward, since it depends on multiple interacting clinical indicators, including hormone measurements and patient medical history. Machine learning provides a useful approach for analyzing complex patient data and identifying patterns that may support diagnostic decision-making. In this project, machine learning models were used to classify thyroid conditions using structured clinical data from the UCI Thyroid Disease dataset.

1.2 Objectives of Project

The main objective of this project was to build and compare machine learning models for thyroid disease classification using patient clinical data. More specifically, the project aimed to preprocess and clean the thyroid dataset, explore the structure and challenges of the data, implement and compare multiple classification models, evaluate model performance using appropriate metrics, analyze the effect of class imbalance on prediction quality, and identify the most effective model and propose improvements for future work.

2. Preprocessing and Exploratory Data Analysis

2.1 Dataset Collection

The dataset used in this project was the Thyroid Disease dataset obtained from the UCI Machine Learning Repository. It contains 9,172 patient records and 29 clinical attributes related to thyroid diagnosis. These features include demographic information, medication indicators, and laboratory measurements such as TSH, T3, TT4, T4U, and FTI. The target variable represents diagnosis labels. In this dataset, the label “-” indicates that no thyroid condition requires comment, while the remaining labels correspond to specific thyroid-related conditions. The dataset is publicly available and appropriate for supervised classification because it contains structured input features and diagnosis labels for each patient.

2.2 Data Pre-Processing

Data pre-processing was conducted in `pre-processing.py` to transform the raw thyroid dataset into a structured format suitable for machine learning. The raw data was first loaded using the documented column schema, with "?" treated as missing values. The script then separated the combined diagnosis field into three components, the full diagnosis label, a `record_id`, and a `diagnosis_primary` variable used as the main target for classification. When multiple diagnoses were recorded in the form X|Y, the script retained the final label as the primary diagnosis. Several cleaning steps were applied to improve data quality and model readiness. Numeric variables were converted to numeric type, and the `tbg` variable was removed because it contained a very large proportion of missing values. Missing values in `sex` were explicitly labeled as "Unknown", while missing values in `referral_source` were replaced with "other". Binary indicator variables were encoded from t and f into 1 and 0, and missing values in selected numeric variables (`age`, `tsh`, `t3`, `tt4`, `t4u`, and `fti`) were imputed using the median of each variable. Finally, categorical variables were one-hot encoded, producing a processed dataset suitable for downstream feature construction and classification modelling.

2.3 Exploratory Data Analysis

Exploratory analysis revealed that the dataset is highly imbalanced. The majority class, represented by "-", accounts for approximately 74 percent of the records, while many other diagnosis classes contain relatively few samples. This imbalance creates a major challenge for classification because models may achieve high overall accuracy simply by favoring the majority class. The dataset also contains both numerical and categorical features, which makes preprocessing especially important. Some laboratory variables contain substantial missing data, and several diagnosis classes have very low support. These characteristics suggest that accuracy alone would not be sufficient for evaluation and that more balanced metrics, such as F1 score, are necessary.

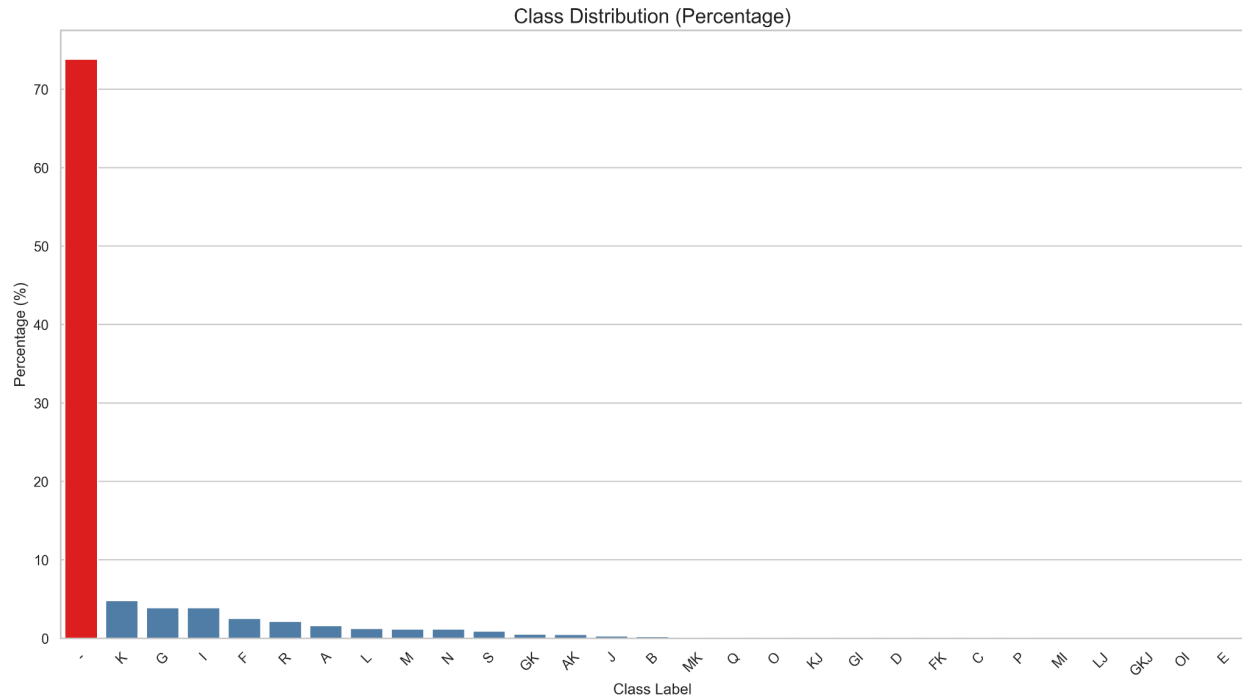


Figure 1: Percentage distribution of diagnosis classes in the thyroid dataset.

As shown in Figure 1, the class distribution is heavily skewed toward the “-” (No diagnosis) label.

2.4 Feature Engineering

The feature engineering stage focused on preparing the preprocessed thyroid dataset for supervised classification by separating predictor variables from the target variable. This step was implemented in `build_features.py` and used the processed dataset as input. Specifically, the script defined `diagnosis_primary` as the target variable to be predicted. The feature matrix was then created by removing `diagnosis_primary`, `diagnosis`, and `record_id` from the dataset. Excluding `diagnosis_primary` was necessary to prevent the target label from being included among the predictors, while `record_id` was removed because it functions only as an identifier and does not provide meaningful information for classification. The `diagnosis` column was also excluded so that the modelling process would rely only on the selected predictive attributes rather than additional diagnosis-related fields. After this separation, the script saved the resulting predictor matrix and target vector as separate files for downstream modelling. This procedure ensured that

the machine learning algorithms were trained using a clearly defined input matrix X and target vector y , providing a consistent structure for model development and evaluation.

3. Methodology

3.1 Platform and Machine Configurations Used

This project was implemented in Python using Visual Studio Code. The workflow included a virtual environment and standard machine learning libraries such as pandas, scikit-learn, matplotlib, seaborn, and joblib. GitHub was used for collaboration.

3.2 Data Split

The data was split into training and testing sets using an 80/20 split. Stratified splitting was used so that class proportions were preserved across the training and testing sets as much as possible.

This is especially important in imbalanced classification tasks because it helps maintain representative class distributions during evaluation.

3.3 Model Planning

Three classification models were selected, Logistic Regression, K-Nearest Neighbours and Random Forest. These models were chosen because they represent different learning strategies. Logistic Regression provides a linear baseline, KNN is a distance-based method, and Random Forest is an ensemble method capable of capturing non-linear relationships. Comparing these models allowed for a broader understanding of how different classification approaches behave on the same dataset.

3.4 Model Training

The model training process used the prepared feature set and corresponding class labels derived from the preprocessed thyroid dataset. Before training, classes with extremely few observations were excluded so that the data could be divided appropriately for supervised learning. The remaining data were then split into training and testing subsets using a stratified approach, ensuring that class proportions were preserved as much as possible across both sets. Three classification models were trained: Logistic Regression, K-Nearest Neighbors, and Random

Forest. Hyperparameter tuning for each model was conducted using cross-validation on the training data. This approach allowed multiple parameter settings to be evaluated systematically before selecting the best-performing configuration for each algorithm. Scaling was applied for Logistic Regression and K-Nearest Neighbors, reflecting the importance of standardized feature values for these methods. In contrast, Random Forest was trained without feature scaling. After model fitting and tuning, each model was evaluated on the held-out test set. The overall training procedure therefore combined stratified data splitting, cross-validated hyperparameter selection, and model-specific preprocessing to support a consistent comparison across the three classification approaches.

3.5 Model Evaluation

Model performance was evaluated using a combination of classification metrics rather than relying on a single measure. The evaluation process included accuracy, precision, recall, and F1 score, together with a confusion matrix. These metrics were necessary because the thyroid classification task is strongly imbalanced, with the non-disease class occurring much more frequently than the other categories. In this setting, accuracy alone is not sufficient. A model can achieve high overall accuracy by predicting the majority class correctly most of the time, while still performing poorly on the less frequent thyroid conditions. For this reason, additional metrics such as precision, recall, and F1 score are needed to assess how well the models distinguish among all classes, not only the dominant one. In particular, class-level and averaged F1-based measures provide a more balanced view of predictive performance under class imbalance.

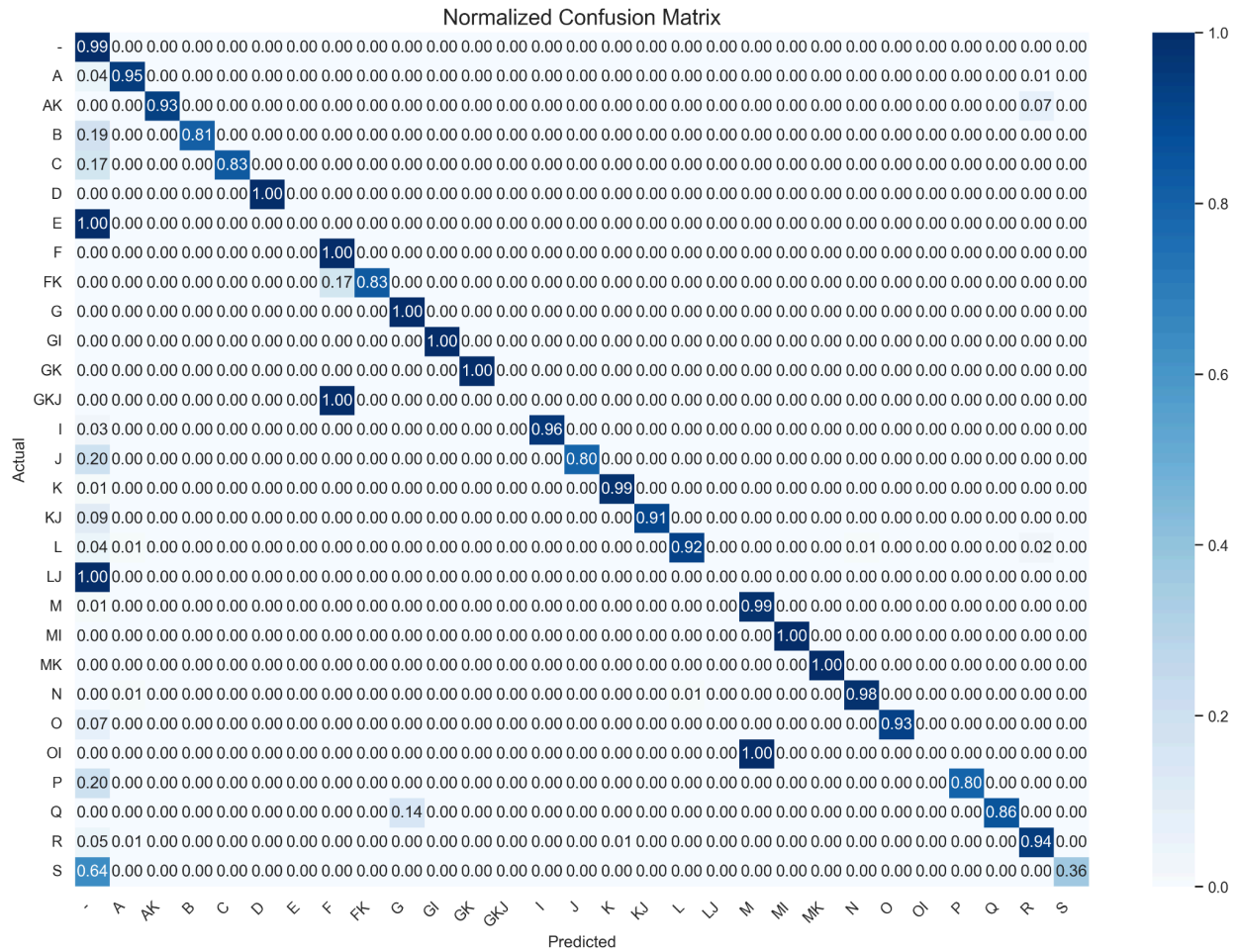


Figure 2: Normalized Confusion Matrix

The confusion matrix complements these summary metrics by showing how predictions are distributed across actual classes, making it easier to identify patterns of misclassification and majority-class bias.

3.6 Model Optimization

Model optimization was implemented through grid-based hyperparameter tuning. Each of the three classification models was trained using a systematic search procedure combined with cross-validation on the training data. The tuning process adjusted different types of parameters depending on the model. For Logistic Regression, optimization focused on the strength of regularization. For K-Nearest Neighbors, the main parameter adjusted was the number of

neighbors used for classification. For Random Forest, the optimization process considered model complexity and ensemble size by varying parameters related to tree depth and the number of trees.

3.7 Final Model Building

Model optimization was performed using hyperparameter tuning combined with cross-validation in order to improve model performance. A Grid Search approach was applied to identify the best combination of parameters for each model based on validation performance. For Logistic Regression, the optimal regularization parameter was found to be $C=10$. For the K-Nearest Neighbors model, the best performance was achieved with $k=3$ neighbors. For the Random Forest model, the optimal configuration consisted of 200 trees with a maximum depth of 20.

These parameters were selected because they produced the highest performance during cross-validation. Despite these optimization efforts, the dataset's severe class imbalance remained a significant challenge. Some classes contained very few samples, which limited the ability of the models to learn meaningful patterns. This resulted in poor performance for some minority classes, even after tuning. The presence of warnings during cross-validation, indicating insufficient samples for certain classes, further highlights this limitation.

4. Results

4.1 Description of the Models

In this study, three different machine learning models were implemented to classify thyroid disease: Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest. Each model represents a different approach to classification and offers unique strengths and limitations. Logistic Regression is a linear classification algorithm that models the probability of a class label based on input features. It is commonly used as a baseline model due to its simplicity and interpretability. In this project, the model was optimized using regularization, with the best parameter identified as $C=10$, which controls the strength of regularization. K-Nearest Neighbors (KNN) is a non-parametric, distance-based algorithm that classifies a data point based on the majority class of its nearest neighbors. The model relies heavily on feature similarity and distance metrics. Through hyperparameter tuning, the optimal number of neighbors was found to be $k=3$.

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs to improve classification performance. It is particularly effective for

handling nonlinear relationships and complex datasets. In this project, the best-performing configuration used 200 trees with a maximum depth of 20, allowing the model to capture detailed patterns in the data while controlling overfitting.

4.2 Performance Metrics

The performance of the models was evaluated using classification metrics, as the task involves predicting categorical labels for different thyroid conditions. Since this is not a regression problem, metrics such as Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) were not applicable. The primary metric used was accuracy, which measures the proportion of correctly classified instances out of the total number of samples. While accuracy provides an overall measure of performance, it can be misleading in cases where the dataset is imbalanced, as a model may achieve high accuracy by simply predicting the majority class. To address this limitation, additional metrics including precision, recall, and F1-score were used. Precision measures the proportion of correctly predicted instances among all instances predicted for a given class, while recall measures the proportion of correctly predicted instances among all actual instances of that class. The F1-score, which is the harmonic mean of precision and recall, provides a balanced measure of a model's performance. In this study, particular emphasis was placed on the macro-averaged F1-score, which calculates the F1-score independently for each class and then takes the average. This metric is especially important in this project due to the significant class imbalance, as it ensures that all classes are treated equally regardless of their frequency. Although metrics such as the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) are commonly used in binary classification problems, they were not used in this study because the task involves multi-class classification with many categories, making AUC-ROC less suitable.

4.3 Results Table

The performance of the three models is summarized in Table 4.1. Both overall accuracy and macro-averaged F1-score are reported to provide a balanced evaluation of model performance, particularly in the presence of class imbalance.

Table 4.3: Performance Comparison of Machine Learning Models

Model	Accuracy	Macro F1 Score	Weighted F1 Score
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Logistic Regression	0.71	0.51	0.75
K-Nearest Neighbors	0.82	0.35	0.80
Random Forest	0.94	0.65	0.93

The results clearly indicate that the Random Forest model outperforms the other models across all evaluation metrics. While K-Nearest Neighbors achieves relatively high accuracy, its low macro F1-score suggests poor performance on minority classes. Logistic Regression provides a more balanced performance than KNN but does not achieve the same level of overall accuracy as Random Forest.

4.4 Interpretation of the Results

The results indicate that the Random Forest model significantly outperforms the other models in terms of both accuracy and overall classification performance. Its ensemble structure allows it to capture complex patterns in the data, making it more effective for this classification task.

Although KNN achieved an accuracy of 0.82, its macro F1-score of 0.35 suggests that it performs poorly on less frequent classes. This indicates that the model is biased toward the majority class and struggles to generalize across all categories.

Logistic Regression, while simpler, demonstrated more balanced performance than KNN, with a macro F1-score of 0.51. However, its overall accuracy was lower, suggesting that it lacks the complexity needed to fully capture the relationships within the dataset.

A detailed examination of the Random Forest classification report further reveals that the model performs very well on the majority class and several well-represented classes, such as F, G, K, and M. However, it completely fails to predict certain minority classes, such as B, C, J, and P, which have precision and recall values of zero. This limitation is primarily due to the severe class imbalance in the dataset.

Additionally, warnings generated during evaluation indicate that some classes had no predicted samples, further confirming that the model struggles with rare categories. This highlights a key

limitation of the current approach and suggests that additional techniques are needed to improve performance on minority classes.

4.5 Visualization

To better understand the performance of the models, several visualizations were generated, including confusion matrices and bar charts of performance metrics. The confusion matrix for the Random Forest model shows a strong concentration along the diagonal, indicating that most predictions are correct. However, off-diagonal values reveal misclassifications, particularly among minority classes. Bar charts comparing accuracy and F1-scores across models clearly illustrate that Random Forest outperforms the other models. Additional visualizations, such as class-wise precision and recall plots, further highlight the imbalance in performance across different classes. These visual tools make it easier to interpret the results and identify areas where the models perform well and where improvements are needed.

4.6 Sensitivity Analysis

A sensitivity analysis was conducted to evaluate how changes in model parameters and data characteristics affect performance. The results show that the models are highly sensitive to class distribution. Classes with a large number of samples are consistently predicted with high accuracy, while those with very few samples are often misclassified or completely ignored.

Hyperparameter tuning improved overall model performance, particularly for the Random Forest model. However, it did not fully address the issue of class imbalance. The presence of warnings during cross-validation, indicating that some classes had fewer samples than the number of splits, further demonstrates the instability of the model when handling rare classes.

This analysis suggests that the robustness of the models could be improved by applying techniques such as resampling, class weighting, or reducing the number of output classes.

5. Conclusion

This project explored the application of machine learning techniques for thyroid disease classification using clinical data. Three models, Logistic Regression, K-Nearest Neighbors, and Random Forest, were implemented and evaluated.

Among the models, Random Forest achieved the best performance, with an accuracy of 94% and a macro F1-score of 0.65. This indicates its strong ability to capture complex patterns in the data. However, the results also revealed a significant limitation related to class imbalance. The models consistently performed poorly on minority classes, which reduces their reliability in real-world medical applications.

The findings highlight the importance of considering both overall accuracy and class-wise performance when evaluating machine learning models, particularly in healthcare contexts where rare cases are critical.

Future work should focus on addressing class imbalance through advanced techniques such as oversampling, undersampling, or synthetic data generation methods such as SMOTE. Additionally, simplifying the classification problem or incorporating more relevant features may further improve model performance.

Overall, this study demonstrates that machine learning has strong potential to support clinical decision-making. However, careful handling of data-related challenges is essential to ensure reliable and fair predictions.